THE COCYCLE DIAGRAM OF A DESCENT DATUM IS A SPLIT EQUALIZER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 2 navigation stubs, for the cocycle condition and for the split equalizer conditions; both identities are named rather than reproduced and the author's use of them is written out.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Editorial note. The label lemma-equalizer-M is defined only in descent.tex, and its canonical tag 08XA is not among the tags of the published items of this file, so it is not a published duplicate.

Allowed standard prerequisites (editorial).

- The category of descent data for modules together with the cocycle condition. Same chapter, the cocycle condition labelled equation-cocycle-condition and the category DD_{S/R} named in the statement at line 1330. Supplies the datum and the identity that makes the splitting compatible.
- Conditions defining a split equalizer. Same chapter, equation-split-equalizer-conditions, invoked at proof line 1356. Supplies the list of compatibilities that the auxiliary morphisms must satisfy.
- Base change of ring homomorphisms along M and the tensor product over S. Standard commutative algebra, used at proof lines 1352-1355. Supplies the auxiliary morphisms of the splitting.

Source reference. Original equation 0: sources/stacks-chow/tags.txt, source lines 1–1. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *For $(M, \theta) \in DD_{S/R}$, the diagram*
(0.1.1)

$$M \xrightarrow{\theta \circ (1_M \otimes \delta_0^1)} M \otimes_{S,\delta_1^1} S_2 \underset{1_{M \otimes S_2} \otimes \delta_1^2}{\overset{(\theta \otimes \delta_2^2) \circ (1_M \otimes \delta_0^2)}{\rightrightarrows}} M \otimes_{S,\delta_{12}^1} S_3$$

is a split equalizer.

Proof. Define the ring homomorphisms $\sigma_0^0 : S_2 \to S_1$ and $\sigma_0^1, \sigma_1^1 : S_3 \to S_2$ by the formulas

$$\sigma_0^0(a_0 \otimes a_1) = a_0 a_1$$
$$\sigma_0^1(a_0 \otimes a_1 \otimes a_2) = a_0 a_1 \otimes a_2$$
$$\sigma_1^1(a_0 \otimes a_1 \otimes a_2) = a_0 \otimes a_1 a_2.$$

We then take the auxiliary morphisms to be $1_M \otimes \sigma_0^0 : M \otimes_{S,\delta_1^1} S_2 \to M$ and $1_M \otimes \sigma_0^1 : M \otimes_{S,\delta_{12}^1} S_3 \to M \otimes_{S,\delta_1^1} S_2$. Of the compatibilities required in (0), the first follows from tensoring the cocycle condition (0) with σ_1^1 and the others are immediate. □